

Black-Box - Top 10 of 30 runs (ranked by Test RMSE)

| Rank | Test RMSE | Val RMSE | Test R2 | Test MAE | Epochs | ES |
|------|-----------|----------|---------|----------|--------|----|
| 1    | 0.0830    | 0.0919   | 0.9210  | 0.0464   | 130    | N  |
| 2    | 0.0843    | 0.0923   | 0.9186  | 0.0474   | 350    | N  |
| 3    | 0.0850    | 0.0936   | 0.9173  | 0.0475   | 416    | N  |
| 4    | 0.0859    | 0.0919   | 0.9156  | 0.0473   | 107    | N  |
| 5    | 0.0862    | 0.0915   | 0.9149  | 0.0468   | 217    | N  |
| 6    | 0.0864    | 0.0928   | 0.9144  | 0.0471   | 158    | N  |
| 7    | 0.0872    | 0.0932   | 0.9129  | 0.0477   | 220    | N  |
| 8    | 0.0877    | 0.0932   | 0.9119  | 0.0478   | 317    | N  |
| 9    | 0.1195    | 0.1355   | 0.8635  | 0.0846   | 350    | N  |
| 10   | 0.1198    | 0.1348   | 0.8630  | 0.0844   | 350    | N  |

Physics-Reg - Top 10 of 230 runs (ranked by Test RMSE)

| Rank | Test RMSE | Val RMSE | Test R2 | Test MAE | Epochs | ES | Phys-W |
|------|-----------|----------|---------|----------|--------|----|--------|
| 1    | 0.0793    | 0.0853   | 0.9280  | 0.0458   | 140    | N  | 0.3    |
| 2    | 0.0798    | 0.0851   | 0.9271  | 0.0483   | 709    | N  | 0.5    |
| 3    | 0.0798    | 0.0870   | 0.9271  | 0.0462   | 599    | N  | 0.3    |
| 4    | 0.0798    | 0.0832   | 0.9270  | 0.0483   | 310    | N  | 0.5    |
| 5    | 0.0802    | 0.0835   | 0.9262  | 0.0484   | 353    | N  | 0.5    |
| 6    | 0.0804    | 0.0827   | 0.9259  | 0.0481   | 198    | N  | 0.5    |
| 7    | 0.0806    | 0.0843   | 0.9255  | 0.0487   | 158    | N  | 0.5    |
| 8    | 0.0808    | 0.0859   | 0.9252  | 0.0461   | 203    | N  | 0.3    |
| 9    | 0.0808    | 0.0845   | 0.9251  | 0.0455   | 307    | N  | 0.3    |
| 10   | 0.0809    | 0.0868   | 0.9250  | 0.0469   | 340    | N  | 0.3    |

Residual-Corr - Top 10 of 134 runs (ranked by Test RMSE)

| Rank | Test RMSE | Val RMSE | Test R2 | Test MAE | Epochs | ES | alpha-Reg |
|------|-----------|----------|---------|----------|--------|----|-----------|
| 1    | 0.0815    | 0.0845   | 0.9238  | 0.0443   | 278    | N  | 0.05      |
| 2    | 0.0816    | 0.0853   | 0.9236  | 0.0453   | 455    | N  | 0.05      |
| 3    | 0.0819    | 0.0863   | 0.9232  | 0.0455   | 435    | N  | 0.05      |
| 4    | 0.0821    | 0.0860   | 0.9228  | 0.0446   | 215    | N  | 0.05      |
| 5    | 0.0824    | 0.0869   | 0.9221  | 0.0458   | 435    | N  | 0.01      |
| 6    | 0.0825    | 0.0862   | 0.9220  | 0.0456   | 394    | N  | 0.01      |
| 7    | 0.0826    | 0.0864   | 0.9218  | 0.0448   | 215    | N  | 0.01      |
| 8    | 0.0827    | 0.0863   | 0.9217  | 0.0457   | 394    | N  | 0         |
| 9    | 0.0828    | 0.0866   | 0.9215  | 0.0449   | 215    | N  | 0         |
| 10   | 0.0828    | 0.0859   | 0.9214  | 0.0452   | 198    | N  | 0.01      |